

# Boran Yesilyurt

# Curriculum Vitae

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## Contact

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## Education

### University of Florida (UF)

Gainesville, USA

*Physics, PhD*

*August 2022 – Expected August 2027*

- Current GPA: 4.00/4.00
- Advisor: Richard P. Woodard
- Teaching Assistant for more than 8 courses
- Referee at Classical and Quantum Gravity (IOP)

### Middle East Technical University (METU)

Ankara, Turkey

*Physics, BSc*

*August 2018 – June 2022*

- Valedictorian-GPA: 4.00/4.00
- Senior Project: *Schwinger Process of Pair Production on Flat Two-Torus*
- Advisor: Seckin Kurkcuoglu

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## Skills

**Programming:** Python, Mathematica, LaTeX, JavaScript(Vanilla), HTML, CSS

**Scientific Computing:** NumPy, SciPy, Matplotlib, symbolic computation, numerical methods

**Quantitative Methods:** Stochastic modeling, Monte Carlo simulation, Perturbation theory, Asymptotic analysis, Statistical mechanics

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## Research Experience

### University of Florida, Physics Department

Advisor: Richard P. Woodard

*Research Assistantship & Doctoral Studies*

*December 2022 – Current*

- Conducting research in quantum field theory, cosmology, and quantum gravity with emphasis on infrared effects and stochastic phenomena in de Sitter spacetime.
- Developed analytical and computational methods for higher-order perturbative calculations in curved spacetime, including dimensional regularization and tensor structure decomposition.
- Independently completed a technically demanding graviton self-energy calculation involving nontrivial symbolic manipulation, pattern recognition and consistency verification across multiple analytical regimes.
- Performed symbolic and numerical computations in Python and Mathematica to validate theoretical predictions and study emergent large-scale stochastic behavior.
- Co-authored six peer-reviewed publications and one first-author manuscript currently in preparation.

### Middle East Technical University, Physics Department

Advisor: Seckin Kurkcuoglu

*Senior Project and Follow-up Research*

*September 2021 – June 2022*

- Studied Schwinger pair production on compact manifolds using analytical quantum field theoretic methods.
- Derived and analyzed equations governing particle creation under nontrivial topological boundary conditions.
- Performed supporting symbolic and numerical calculations to test theoretical consistency and limiting behavior.

### Middle East Technical University, Physics Department

*Independent Research*

*2018 – 2019*

- Developed a generalized equations-of-motion framework for multi-body pendulum systems.
- Formulated and solved coupled nonlinear dynamical equations for constrained classical systems.

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## Publications

Order of the authors in HEP-TH follows last name alphabetical order. Exceptions are listed with an Asterisk.

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|-------------------------------------------------------------------------------------------------|----------------------------------------|
| Yesilyurt, Tsamis, Woodard - Dimensionally Regulated Graviton Self-Energy from Gravitons on dS* | Manuscript in Preparation              |
| Woodard, Yesilyurt - The Other ADM                                                              | J.Phys.A 58 (2025) 19, 195401          |
| Miao, Tsamis, Woodard, Yesilyurt - The Third Structure Function                                 | Class.Quant.Grav. 42 (2025) 10, 105001 |
| Tsamis, Woodard, Yesilyurt - The Need to Renormalize Cosmological Constant                      | Phys.Lett.B 849 (2024) 138472          |
| Litos, Woodard, Yesilyurt - Large Logs in a Nontrivial Nonlinear Sigma Model                    | Phys.Rev.D 108 (2023) 6                |
| Woodard, Yesilyurt - Remembrance of Things Past                                                 | JHEP 08 (2023) 124                     |
| Woodard, Yesilyurt - Unfinished Business in a Nonlinear Sigma Model on dS Background            | JHEP 06 (2023) 206                     |
| Yesilyurt - Equations of Motion of A Pendulum Containing N-point Masses                         | arXiv:1910.12610 (2019)                |

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## Projects

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Stochastic Modeling of Bacterial Colony Growth</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Python, stochastic simulation, data analysis |
| <ul style="list-style-type: none"><li>Developed a phenomenological model for effective colony-size growth using symmetry-based arguments, stochastic dynamics and local interaction dynamics</li><li>Implementing biased random-walk and Monte Carlo methods to study emergent growth behavior and compare model predictions across parameter regimes by simulating multiple independent realizations</li><li>Computed statistical observables across runs, including mean growth profiles and variance evolution over time</li><li>Analyzed how stochastic fluctuations affect macroscopic colony structure formation</li><li>Built visualization tools to track spatial configuration and temporal evolution of colony states across iterations</li><li>Building a reproducible research notebook with structured visualizations and documented future directions</li></ul> |                                              |
| <b>Interactive Elden Ring Companion App</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Python, JavaScript, HTML, CSS                |
| <ul style="list-style-type: none"><li>Designed a companion utility tool for organizing and accessing structured game information providing an interactive game data exploration experience</li><li>Implemented dynamic UI logic and state management using Vanilla JavaScript</li><li>Implemented data aggregation and lookup system for in-game items, builds, and mechanics</li><li>Built modular architecture to allow easy extension of content categories and future updates</li><li>Improved user experience by centralizing dispersed game information into a single interface</li></ul>                                                                                                                                                                                                                                                                               |                                              |

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## Scholarships & Awards

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|--------------------------------------------------------|---------------------------|
| Institute of Fundamental Theory(IFT) Summer Fellowship | May - August 2025, UF     |
| DGRAV APS 2025 Travel Grant                            | 2025, Anaheim, CA         |
| Charles F. Hooper Jr. Memorial Award                   | 2024, UF                  |
| TRISEP 2023 Travel Grant                               | 2023, Perimeter Institute |
| Teaching Fellowship                                    | 2022-Current, UF          |
| TÜBİTAK - 2205 Scholarship                             | 2017-2022, METU           |
| Yükseköğretim Kurulu (YÖK) Destek Scholarship          | 2021-2022, METU           |

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## Talks

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|----------------------------------------------------------------------|--------------------------------------------|
| 2025 APS Joint April & March Meeting - Poster Presentation           | 16-21 March, 2025, Anaheim, CA             |
| 10th Gulf Coast Gravity Meeting - Talk                               | 1-2 March, 2024, UT-Austin, TX             |
| The Early Universe: A Window to New Physics - Talk                   | 20-22 October, 2023, University of Florida |
| TRISEP 2023 Summer School, Perimeter Institute - Poster Presentation | 19-30 June, 2023, Ontario/Canada           |

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## Teaching

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|-------------------------------------------------------|-----------------------|
| Physics - 1: Labs                                     | University of Florida |
| Physics - 1: Discussion Teaching Assistant(TA)        | University of Florida |
| PHZ3113 Intro to Theoretical Physics: TA              | University of Florida |
| PHY4324 Electromagnetism 2: TA                        | University of Florida |
| PHY4604 Quantum Mechanics 1: TA                       | University of Florida |
| PHY6347 (Graduate Level) Electromagnetic Theory 2: TA | University of Florida |
| PHY6646 (Graduate Level) Quantum Mechanics 2: TA      | University of Florida |

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## ***Service, Outreach & Event Organization***

### **Referee**

*Classical and Quantum Gravity (IOP)*

### **Graduate Students and Postdoc Seminars (GSPS)**

UF

*Monthly HEP-TH Seminar Organizer*

*September 2023 – Ongoing*

### **The 62nd Florida Regional Junior Science and Humanities Symposium**

UF

*Referee for Admission to STEM Research Speaker Competition*

*2-4 February, 2025*

### **Physics Society**

METU

*Board Member between March 2018-2019 September*

*October 2017 – October 2019*

- Demonstrated experiments in Yenimahalle Science Fair- 2019, Ankara
- Organized Interscholastic Physics Students Summit (IPSS)- 2018 and IPSS- 2019
- Coordinated and organized weekly seminars with lecturers from Physics department on topics out of the curriculum
- Attended International Conference of Physics Students (ICPS)- 2019 held at Cologne, Germany as representative and participant

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## **REFERENCES**

- Available upon request.